# IT SERVICE DESK & INCIDENT ANALYTICS

## Complete Project Documentation

### Project Purpose

A business-focused Excel analytics solution designed to evaluate IT service workload, SLA performance, asset risk, incident impact, technician capacity, and customer experience through structured PivotTables, PivotCharts, derived fields, conditional formatting, KPI analysis, and an executive management dashboard.

| Component | Scope |
|---|---|
| Business Need | Operational visibility, service performance, risk identification, and management decision support |
| Primary Platform | Microsoft Excel |
| Analytical Methods | PivotTables, PivotCharts, counts, averages, sums, cross-tabs, bands/flags, conditional formatting |
| Datasets | Eight datasets in the workbook; seven are explicitly identified in the documented analysis: Users, Technicians, Tickets, Assets, SLA, Feedback, Incidents, plus an eighth workbook dataset |
| Final Deliverable | IT Service Desk & Incident Analytics Dashboard supported by detailed analytical PivotTables |

## 1. Executive Summary

This project developed an Excel-based IT Service Desk & Incident Analytics (ITSM) analytics solution around a management need: turn operational IT records into a clear view of workload, service quality, operational risk, incident impact, support capacity, and customer experience.

The work progressed from dataset/field validation through structured PivotTables, band and flag analysis, cross-dimensional matrices, selected PivotCharts, KPI consolidation, and a management dashboard. Detailed analysis remains available in supporting sheets while the dashboard concentrates the most decision-useful measures.

### Headline Results

| Area | KPI | Result |
|---|---|---|
| Tickets | Total Tickets | 7,000 |
| Tickets | SLA Compliance | 98.94% |
| Tickets | Avg First Response | 1.57 hrs |
| Tickets | Avg Resolution | 4.96 hrs |
| Tickets | SLA Breaches | 74 |
| Tickets | Escalations | 827 |
| Tickets | Reopened Tickets | 556 |
| Assets | Total Assets | 1,500 |
| Assets | Warranty Risk Assets | 1,153 (76.9%) |
| Assets | Expired Warranty Assets | 1,106 |
| Incidents | Total Incidents | 1,200 |
| Incidents | Open Incidents | 291 |
| Incidents | 361+ Minute Downtime | 570 (47.5%) |
| Incidents | 1,001+ Users Affected | 716 (59.7%) |
| SLA | SLA Records | 600 |
| SLA | Avg Response Target | 9.27 hrs |
| SLA | Avg Resolution Target | 37.56 hrs |
| SLA | Avg Escalation Target | 53.24 hrs |
| Feedback | Surveys | 3,000 |
| Feedback | Avg Rating | 3.78 / 5 |
| Feedback | Recommendation | 75.1% |
| Feedback | Positive Indicators | 1,912 |
| Feedback | Negative Indicators | 487 |

### Business Interpretation

- Service operations handle substantial demand while maintaining strong overall SLA compliance.
- Warranty exposure is a major asset-management risk: 1,153 of 1,500 assets are warranty risk.
- Long-duration incidents are material: 570 incidents fall into the 361+ minute downtime band.
- Incident reach is broad: 716 incidents affected 1,001+ users.
- Customer sentiment is generally positive, with a 3.78/5 average rating and 75.1% recommendation rate.

## 2. Business Need and Objectives

The project was structured around the practical needs of an IT service organization. Management needs summarized indicators-not raw records alone-to understand demand, handling performance, SLA commitments, asset exposure, incident disruption, support capacity, and customer perception.

### Core Business Questions

- How much IT service demand is being handled and where is ticket volume concentrated?
- How quickly are tickets answered and resolved?
- Where are SLA breaches, escalations, and reopened tickets concentrated?
- Which departments, categories, locations, priorities, and statuses create the greatest workload?
- How large and costly is the asset base, and where are warranty and repair risks concentrated?
- Which incident types and root causes create the greatest impact and downtime?
- Which services and responsible teams experience the highest incident exposure?
- How much of the incident portfolio is open, unresolved, or critical-impact?
- What SLA targets exist across priority, severity, service, ticket type, category, and SLA level?
- How satisfied are customers and how likely are they to recommend the service?

### Project Objectives

1. Build a structured analytical view of the IT service operation.
2. Use actual available fields and avoid inventing unavailable dimensions.
3. Create repeatable PivotTable analyses for workload, performance, risk, and experience.
4. Use conditional formatting to reveal relative concentration and risk.
5. Create selected PivotCharts for executive communication.
6. Consolidate the strongest KPIs into a single management dashboard.
7. Document the workflow for portfolio and professional presentation.

## 3. Data Architecture and Validation

The workbook contains eight datasets. Seven were explicitly identified during the documented analysis: Users, Technicians, Tickets, Assets, SLA, Feedback, and Incidents. The eighth dataset is confirmed to exist by the project workflow, but its name and field structure were not established in the available record; it is therefore not named or assigned unsupported content in this document.

| Dataset | Purpose | Main Themes |
|---|---|---|
| Users | User/workforce profile | Department, location, employment type/status, hire year, tenure |
| Technicians | Support capacity profile | Specialization, location, experience, shift, status, tenure |
| Tickets | Service-desk workload/performance | Volume, status, category, priority, department, location, technician, SLA, response, resolution, escalation, reopening |
| Assets | Inventory/lifecycle/risk | Type, manufacturer, department, status, warranty, age, replacement cost, risk |

| | | |
|---|---|---|
| SLA | Service commitments | Priority, severity, category, ticket type, service, SLA level, strictness, target bands |
| Feedback | Customer experience | Rating, rating band, recommendation, positive/negative indicators |
| Incidents | Service disruption/cause | Type, root cause, impact, service, users, downtime, team, status, flags/bands |
| Eighth workbook dataset | Part of eight-dataset structure | Name/fields not established in documented analysis |

### Field Integrity Decisions

- Experience Level was confirmed to belong to the Technicians table, not Users.
- Assets were confirmed to use Replacement Cost for financial exposure; no generic Cost field was assumed.
- The SLA table was confirmed not to contain Channel, so no Channel analysis was invented.
- Existing derived fields such as Tenure_Band, Response_Target_Band, Resolution_Target_Band, Impact_Band, Downtime_Band, affected_Users_Band, and operational flags were used where available.

**Analytical Principle**

Use the source fields that actually exist, validate their table ownership, and avoid filling gaps with assumptions.

## 4. Analytical Methodology

- Counts were used for workload, population, and record-volume measures.
- Averages were used for response time, resolution time, target hours, downtime, affected users, and tenure.
- Sums were used for binary/event indicators such as SLA breaches, escalations, reopened tickets, and feedback indicators.
- Cross-tab PivotTables were used to identify concentrations across dimensions.
- Conditional color scales were used on comparative matrices.
- Charts were selected according to the question: bars/columns for comparison, lines for trends, and doughnuts for simple composition.
- The executive dashboard intentionally contains a subset of the detailed analytical charts to avoid overcrowding.

## 5. Users Analysis

Users analysis established the organizational context behind service demand.

| Analysis | Result |
|---|---|
| Users by Department | Engineering 122; HR 117; Legal 104; Procurement 101; Operations 100; Customer Service 100; IT 86. |
| Users by Location | Makkah 170; Madinah 165; Taif 158; Khobar 153; Riyadh 150; Jeddah 138; Abha 133; Dammam 133. |
| Employment Type | Contract 33%; Full-Time 35%; Part-Time 32%. |
| Employment Status | Active 90%; Inactive 10%. |
| Avg Tenure by Department | Customer Service 4.91 years highest; Logistics 4.39 lowest; overall 4.61 years. |
| Hire Year | 2018 160; 2019 134; 2020 160; 2021 136; 2022 150; 2023 134; 2024 151; 2025 142; 2026 33. |

## 6. Technicians Analysis

Technician analysis measured support capacity and workforce distribution.

| Analysis | Result |
|---|---|
| Specialization | Applications 87; Cybersecurity 83; Database 81; Networking 80; Windows 74; Cloud 68; Microsoft 365 65; Hardware 62. These eight specializations were retained. |
| Location | Makkah 89; Khobar 88; Abha 77; Jeddah 74; Taif 72; Dammam 69; Madinah 69; Riyadh 62. |
| Experience | Junior 162; Lead 147; Mid-Level 146; Senior 145. |
| Shift | Day 37%; Night 33%; Evening 30%. |
| Employment Status | Active 83%; On Leave 9%; Inactive 8%. |
| Avg Tenure by Experience | Junior 5.05; Lead 5.20; Mid-Level 5.25; Senior 5.01; overall 5.124 years. |
| Active by Experience | Junior 143; Lead 118; Mid-Level 123; Senior 117; total active 501. |
| Active by Shift | Day 184; Night 164; Evening 153; total active 501. |
| Tenure Band | 5–10 years 314; 1–3 years 132; 3–5 years 126; <1 year 28. |

## 7. Tickets Analysis

Tickets were treated as the central service-desk workload dataset.

| Core Metric | Result |
|---|---|
| Ticket Volume | 7,000 |
| Avg First Response | 1.57 hrs |
| Avg Resolution | 4.96 hrs |
| SLA Breaches | 74 |
| Escalations | 827 |

| | |
|---|---|
| Reopened Tickets | 556 |

## Ticket Status

| Status | Tickets |
|---|---|
| Cancelled | 283 |
| Closed | 2,092 |
| In Progress | 863 |
| Open | 565 |
| Pending User | 520 |
| Resolved | 2,677 |
| Total | 7,000 |

Open/Active Tickets was defined in the project as In Progress + Open + Pending User = 1,948.

## Ticket Volume by Category

| Category | Tickets |
|---|---|
| Hardware | 1,401 |
| Network | 1,289 |
| Software | 1,266 |
| Access | 1,201 |
| Email | 988 |
| Security | 855 |
| Total | 7,000 |

## Department Performance

| Department | Vol. | Resp. | Res. | Breaches | Escalations | Reopened |
|---|---|---|---|---|---|---|
| HR | 706 | 1.53 | 4.89 | 4 | 70 | 58 |
| Engineering | 673 | 1.55 | 4.89 | 10 | 68 | 56 |
| Legal | 615 | 1.60 | 4.83 | 9 | 75 | 53 |
| Procurement | 612 | 1.65 | 5.06 | 9 | 82 | 45 |
| Logistics | 598 | 1.63 | 4.83 | 3 | 83 | 46 |
| Operations | 588 | 1.60 | 5.29 | 8 | 63 | 49 |
| Customer Service | 584 | 1.59 | 4.99 | 8 | 78 | 42 |
| Finance | 566 | 1.53 | 5.03 | 4 | 64 | 38 |
| Administration | 540 | 1.54 | 5.06 | 7 | 65 | 54 |
| Marketing | 525 | 1.53 | 5.05 | 7 | 61 | 42 |
| Sales | 520 | 1.55 | 4.84 | 1 | 65 | 39 |
| IT | 473 | 1.54 | 4.76 | 4 | 53 | 34 |

## Other Ticket Views

- Top 10 technicians by volume: TECH0366 led with 24 tickets; several others were at 20–21.
- Location volume: Madinah 1,000; Makkah 973; Khobar 907; Taif 873; Riyadh 869; Abha 826; Jeddah 796; Dammam 756.

- Ticket status by department was analyzed to locate operational workload.
- Priority/category/location/technician views were used for deeper workload and service-performance investigation.
- Monthly ticket volume was charted across 2025–2026.

## 8. Assets Analysis

Asset analysis converted the asset register into an inventory, lifecycle, cost, and risk view.

| Asset KPI | Result |
|---|---|
| Total Assets | 1,500 |
| Total Replacement Cost | 11,450,698.61 |
| Warranty Risk Assets | 1,153 |
| Expired Warranty Assets | 1,106 |
| Assets Under Repair | 391 |

### Asset Type

| Type | Count |
|---|---|
| Firewall | 163 |
| Printer | 159 |
| Laptop | 158 |
| Desktop | 155 |
| Access Point | 152 |
| Monitor | 148 |
| Mobile Device | 148 |
| Switch | 147 |
| Server | 145 |
| Router | 125 |

### Asset Status and Replacement Cost

| Status | Count | Replacement Cost | Avg Replacement Cost |
|---|---|---|---|
| Under Repair | 391 | 3,043,326.05 | 7,783.44 |
| In Use | 375 | 2,908,539.56 | 7,756.11 |
| Retired | 385 | 2,815,098.74 | 7,311.94 |
| In Stock | 349 | 2,683,734.26 | 7,698.78 |
| Total | 1,500 | 11,450,698.61 | 7,633.80 |

### Warranty Status

| Warranty | Count | Replacement Cost |
|---|---|---|
| Expired | 1,106 | 8,425,687.19 |
| Active | 347 | 2,627,173.56 |
| Expiring Within 90 Days | 47 | 397,837.86 |
| Total | 1,500 | 11,450,698.61 |

### Asset Age and Risk Views

| Age Band | Assets |
|---|---|
| <1 Year | 193 |
| 1–3 Years | 383 |
| 3–5 Years | 378 |
| 5–7 Years | 404 |
| 7+ Years | 142 |
| Total | 1,500 |

- Warranty risk was analyzed by department, asset type, location, status, and age.
- IT had 118 warranty-risk assets out of 146; Dammam had 166 out of 204; Firewall had 129 out of 163.
- Assets by manufacturer were also analyzed: HP 178; Samsung 154; Fortinet 151; Cisco 150; Dell/Epson 148 each; Microsoft 147; Lenovo/Huawei 142 each; Apple 140.

## 9. SLA Analysis

SLA analysis established the service-commitment layer and explicitly used only fields present in the SLA table.

| Dimension | Key Results |
|---|---|
| Priority | Critical 162; High 146; Low 155; Medium 137. |
| Severity | High 158; Low 157; Critical 150; Moderate 135. |
| Ticket Type | Service Request 132; Problem 127; Change Request 124; Incident 109; Access Request 108. |
| SLA Level | Critical Support 220; Standard 195; Premium 185. |
| Overall | 600 SLA records; response 9.27 hrs; resolution 37.56 hrs; escalation 53.24 hrs. |

### Target Structure

| View | Key Results |
|---|---|
| Service | ERP response 7.66, resolution 32.00, escalation 45.53; CRM response 6.69; VPN response 12.54, resolution 47.41, escalation 66.44. |
| Severity | Critical: 8.13 / 34.51 / 49.28 hrs; Low: 10.27 / 40.97 / 57.94 hrs. |
| SLA Level | Critical Support 9.30 / 38.29 / 54.44; Standard 8.63 / 35.41 / 50.28; Premium 9.92 / 38.96 / 54.94. |
| Response Bands | ≤2 162; 3–4 146; 5–8 137; 9+ 155. |
| Resolution Bands | ≤8 162; 9–24 146; 25–48 137; 49+ 155. |
| Strictness | High 185; Standard 195; Strict 220; target gap 29.04, 26.78, 29.00 hrs respectively. |

- Ticket Type × Priority matrix was created as a supporting analytical view.
- Category × Severity matrix was created as a supporting analytical view.
- SLA by Service was created as a supporting analytical view.

- SLA by Ticket Type was created as a supporting analytical view.
- SLA by Severity was created as a supporting analytical view.
- SLA by SLA Level was created as a supporting analytical view.

## 10. Feedback Analysis

Feedback analysis focused on customer satisfaction, recommendation behavior, and positive/negative indicators.

| Rating | Surveys |
|---|---|
| 1.00 | 43 |
| 1.33 | 91 |
| 1.67 | 79 |
| 2.00 | 93 |
| 2.33 | 103 |
| 2.67 | 142 |
| 3.00 | 193 |
| 3.33 | 241 |
| 3.67 | 293 |
| 4.00 | 332 |
| 4.33 | 477 |
| 4.67 | 567 |
| 5.00 | 346 |
| Total | 3,000 |

### Rating Bands

| Band | Surveys | Avg Rating |
|---|---|---|
| High | 2,083 | 4.33 |
| Neutral | 538 | 3.03 |
| Low | 379 | 1.78 |
| Total | 3,000 | 3.78 |

### Recommendation and Feedback

| Measure | Result |
|---|---|
| Recommend Yes | 2,253 (75.1%) |
| Recommend No | 747 (24.9%) |
| Positive Feedback indicators | 1,912 |
| Negative Feedback indicators | 487 |

- Rating × Recommendation and Rating Band × Recommendation were analyzed.
- High: 1,574 Yes / 509 No; Neutral: 395 Yes / 143 No; Low: 284 Yes / 95 No.
- Rating Band × Feedback: High 1,329 positive / 324 negative; Neutral 350 / 90; Low 233 / 73.
- Positive_Feedback and Negative_Feedback were treated as separate indicators, not mutually exclusive outcomes.

## 11. Incident Analysis

Incident analysis examined disruption type, causes, impact, affected services/users, downtime, responsible teams, status, and operational flags.

| Incident Type | Count |
|---|---|
| Major Incident | 259 |
| Performance Incident | 244 |
| Security Incident | 236 |
| Service Outage | 231 |
| Recurring Incident | 230 |
| Total | 1,200 |

### Root Causes

| Root Cause | Count |
|---|---|
| Capacity Issue | 138 |
| User Error | 130 |
| Third-Party Failure | 130 |
| Configuration Error | 126 |
| Software Bug | 119 |
| Hardware Failure | 119 |
| Power Failure | 119 |
| Security Event | 111 |
| Unknown | 107 |
| Network Outage | 101 |

### Impact, Service and Status

| View | Key Result |
|---|---|
| Impact | Critical 324; High 301; Medium 296; Low 279. |
| Affected users | Average 1,232.53; 716 incidents affected 1,001+ users. |
| Downtime | Average 5.86 hrs; 570 incidents at 361+ minutes. |
| Service | Email 139 incidents; Network 136; Internet/ERP 134 each; VPN 91. |
| Highest service impact | VPN avg affected users 1,349.96; Internet avg downtime 6.20 hrs. |
| Status | Closed 299; Investigating 281; Open 291; Resolved 329. |

### Incident Bands and Flags

| Analysis | Result |
|---|---|
| Downtime Band | 0–30: 45; 31–120: 154; 121–360: 431; 361+: 570. |
| Affected Users Band | 1–50: 27; 51–250: 101; 251–1,000: 356; 1,001+: 716. |
| Critical Impact Flag | 0: 876; 1: 324. |
| Open Incident Flag | 0: 909; 1: 291. |
| Resolved Incident Flag | 0: 871; 1: 329. |

| | |
|---|---|
| Responsible Team | Desktop Support 187; Applications 176; Database/Infrastructure/Security 175 each; Network/Service Desk 156 each. |

### Cross-Analysis Findings

- Incident Type × Impact: Performance Incident had the highest Critical count at 72.
- Root Cause × Impact: User Error had 40 Critical-impact incidents; Hardware Failure had 40 High-impact incidents.
- Root Cause × Downtime: User Error had 67 incidents at 361+ minutes; Third-Party Failure 65; Hardware Failure 63.
- Affected Service × Downtime: Internet had the highest 361+ minute count at 73.
- Root Cause × Service: Third-Party Failure + Network had 23 incidents; User Error + Network had 21.
- Responsible Team: Security had the highest average affected users at 1,308.36; Applications had the highest average downtime at 6.20 hrs.

## 12. Executive Dashboard

The final dashboard was designed as an executive summary rather than a duplicate of every analytical PivotTable. Its hierarchy is workload → service performance → asset risk → incidents → customer experience.

### Dashboard KPI Cards

| KPI | Value |
|---|---|
| Total Tickets | 7,000 |
| SLA Compliance | 98.94% |
| Avg Resolution | 4.96 hrs |
| SLA Breaches | 74 |
| Open/Active Tickets | 1,948 |
| Reopened Tickets | 556 |
| Total Assets | 1,500 |
| Warranty Risk Assets | 1,153 |
| Open Incidents | 291 |
| Long-Downtime Incidents | 570 |
| Avg Customer Rating | 3.78 / 5 |
| Recommendation Rate | 75.1% |

### Dashboard Charts

- Tickets by Category
- Tickets by Status
- SLA Compliance by Priority
- Monthly Ticket Volume Trend
- Assets by Status
- Incidents by Downtime Band
- Customer Ratings Distribution
- Customer Recommendation

- Average Response & Resolution Time by Priority

### Management Highlights

- Asset Warranty Risk - 1,153 of 1,500 assets (76.9%) are warranty risk, including 1,106 expired assets.
- Incident Downtime - 570 of 1,200 incidents (47.5%) experienced 361+ minutes of downtime.
- Incident Exposure - 716 incidents affected 1,001+ users (59.7%).
- SLA Performance - 7,000 tickets achieved 98.94% SLA compliance, with 74 breaches.
- Customer Experience - Average rating is 3.78/5 and 75.1% would recommend the service.

## 13. Key Business Insights

i. **Service workload:** Hardware and Network are the largest ticket categories, at 1,401 and 1,289 respectively.
ii. **SLA:** Compliance is high at 98.94%, but 827 escalations and 556 reopened tickets indicate additional operational workload.
iii. **Active workload:** 1,948 tickets are Open/Active under the project definition.
iv. **Asset risk:** Warranty risk is 1,153 assets (76.9%), including 1,106 expired warranties.
v. **Financial exposure:** Total replacement cost is 11.45 million; Under Repair assets represent 3.04 million.
vi. **Incident duration:** 47.5% of incidents fall into the 361+ minute downtime band.
vii. **Incident reach:** 59.7% of incidents affect 1,001+ users.
viii. **Root causes:** Capacity Issue is the largest root-cause group; User Error and Third-Party Failure are also important and contribute materially to prolonged downtime.
ix. **Service exposure:** Internet has the highest average incident downtime; VPN has the highest average affected-user count.
x. **Customer experience:** Average rating is 3.78/5 and recommendation is 75.1%.

## 14. Business Recommendations

i. **Prioritize warranty remediation:** Build a renewal/replacement plan for expired and soon-to-expire assets, prioritizing high-risk assets and departments.
ii. **Reduce prolonged incidents:** Target the 361+ minute population and focus corrective actions on User Error, Third-Party Failure, Hardware Failure, and Capacity Issue.
iii. **Strengthen problem management:** Use root-cause × impact and root-cause × downtime matrices to prioritize preventive actions.
iv. **Monitor active ticket workload:** Track the 1,948 Open/Active tickets and segment them by department, priority, category, and status.
v. **Review escalation/reopen drivers:** Investigate the 827 escalations and 556 reopened tickets for handoff, knowledge, resolution, or process issues.
vi. **Protect high-impact services:** Prioritize Internet, ERP, Email, and Network patterns where frequency, affected users, or downtime is material.
vii. **Use customer feedback:** Investigate the 25% non-recommendation segment and negative feedback indicators for recurring dissatisfaction drivers.

viii. **Maintain SLA governance:** Compare actual handling performance with SLA targets by priority, severity, service, and ticket type.

ix. **Maintain reporting governance:** Refresh PivotTables and dashboard elements on a controlled cadence while retaining detailed analysis for auditability.

## 15. Tools, Techniques and Portfolio Skills

| Area | Skills Demonstrated |
|---|---|
| Excel Analytics | PivotTables, PivotCharts, filtering, sorting, derived fields, conditional formatting |
| Data Analysis | Counts, averages, sums, segmentation, trend analysis, cross-tab analysis |
| ITSM Analytics | Tickets, SLA, incidents, assets, technicians, users, customer feedback |
| KPI Design | SLA compliance, workload, response/resolution, risk, downtime, satisfaction |
| Dashboarding | KPI cards, chart selection, layout hierarchy, management highlights |
| Business Communication | Findings, risk statements, recommendations, executive storytelling |

## 16. Project Deliverables

- Validated source-data structure and field ownership.
- Detailed PivotTables across the eight-dataset workbook workflow.
- PivotCharts for selected operational, risk, incident, and customer views.
- Conditional-formatting matrices for comparative analysis.
- Executive IT Service Desk & Incident Analytics Dashboard.
- Management Highlights section.
- Complete project documentation.

## 17. Quality Assurance

| QA Check | Status |
|---|---|
| Dataset fields validated before analysis | Completed |
| Unavailable fields avoided | Completed |
| Experience Level treated as Technician field | Completed |
| Replacement Cost used for asset financial analysis | Completed |
| SLA Channel not assumed | Completed |
| Existing bands/flags reused | Completed |
| PivotTable totals checked | Completed |
| Dashboard KPI values aligned to project results | Completed |
| Detailed analysis retained outside dashboard | Completed |

## 18. Scope Notes and Limitations

- The workbook contains eight datasets, but only seven dataset names were explicitly established in the documented analysis. The eighth is not named here to avoid unsupported invention.
- The final dashboard state reflects the approved project direction. Three previously discussed cleanup items were intentionally left unchanged: the SLA compliance chart presentation, the detailed warranty table/chart choice, and the August 2026 ticket-trend point.
- Positive_Feedback and Negative_Feedback are independent indicators and were not treated as mutually exclusive outcomes.
- The project is descriptive analytics: it identifies patterns and priorities but does not claim causality where the data supports only association.
- Open/Active Tickets is defined as In Progress + Open + Pending User.
- Replacement Cost is the asset financial measure because no generic Cost field was available.

## 19. Conclusion

The completed IT Service Desk & Incident Analytics Analytics project transforms a multi-dataset operational environment into a structured management reporting solution. It covers the user population, technician capacity, ticket workload, asset lifecycle and financial exposure, SLA commitments, customer feedback, and incident performance.

The strongest management signals are clear: workload is substantial, SLA compliance is strong, asset warranty risk is significant, prolonged incident downtime is material, incident reach can be broad, and customer sentiment is generally positive. The dashboard consolidates these signals while the supporting PivotTables preserves the detail needed for operational investigation.

**Portfolio Positioning**

This project demonstrates an end-to-end Excel analytics workflow: business-question framing → data/field validation → PivotTable analysis → cross-dimensional investigation → KPI selection → executive dashboard → management insights and recommendations.

### Appendix A - Analytical PivotTable / Chart Inventory

- Users by Department; Users by Location; Users by Employment Type; Users by Employment Status; Users by Hire Year; Average Tenure by Department.
- Technicians by Specialization; Location; Experience Level; Shift; Employment Status; Average Tenure by Experience; Active Technicians by Experience; Active Technicians by Shift; Tenure Band.
- Tickets by Category; Tickets by Status; Ticket Performance by Department; Ticket Volume by Location; Top 10 Technicians by Ticket Volume; Monthly Ticket Volume Trend; Ticket Status × Department.
- Assets by Type; Manufacturer; Department and Replacement Cost; Status; Warranty Status; Age; Warranty Risk by Department/Type/Location.
- SLA by Priority; Severity; Ticket Type; Service; Response Target Band; Resolution Target Band; SLA Level; SLA Strictness/Target Gap; Ticket Type × Priority; Category × Severity.

- Feedback Ratings Distribution; Rating Band; Customer Recommendation; Positive vs Negative Feedback; Rating Band × Feedback; Rating × Recommendation; Rating Band × Recommendation.
- Incidents by Type; Root Cause; Impact Level; Affected Service; Responsible Team; Status; Downtime Band; Affected Users Band; Critical/Open/Resolved Flags; Incident Type × Impact; Root Cause × Impact; Root Cause × Downtime; Affected Service × Downtime; Root Cause × Affected Service.

## Appendix B - Dashboard KPI Definitions

| KPI | Definition |
|---|---|
| Total Tickets | Count of ticket records = 7,000. |
| SLA Compliance | Overall project result = 98.94%. |
| Avg Resolution | Average ticket resolution time = 4.96 hrs. |
| SLA Breaches | Total SLA breach flags = 74. |
| Open/Active Tickets | In Progress + Open + Pending User = 1,948. |
| Reopened Tickets | Total reopen flags = 556. |
| Total Assets | Asset records = 1,500. |
| Warranty Risk Assets | Warranty-risk assets = 1,153. |
| Open Incidents | Open Incident Flag = 1 = 291. |
| Long-Downtime Incidents | 361+ minute downtime band = 570. |
| Avg Customer Rating | Average Overall Rating = 3.78/5. |
| Recommendation Rate | Yes recommendations 2,253 / 3,000 = 75.1%. |